

Patrick FERNANDES

PERSONAL DATA

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ACADEMIC QUALIFICATIONS

Current Doctor of Philosophy in LANGUAGE TECHNOLOGIES
Carnegie Mellon University, Pittsburgh
Doctor of Philosophy in ELECTRICAL AND COMPUTER ENGINEERING
Instituto Superior Técnico, Lisbon

JUNE 2018 Master of Philosophy in ADVANCED COMPUTER SCIENCE
University of Cambridge, Cambridge
Distinction

JULY 2017 Bachelor of Science in COMPUTER ENGINEERING
Instituto Superior Técnico, Lisbon
Average Grade: 19/20
Graduated 1st out of class of 170

JULY 2014 High School Degree in SCIENCE
Escola Secundária "Ferreira Dias", Sintra
Average Grade: 18/20
Graduated 1st out of class of 1000+

PUBLICATIONS

Published | MEASURING AND INCREASING CONTEXT USAGE IN
CONTEXT-AWARE MACHINE TRANSLATION
P. Fernandes*, K. Yin, G. Neubig, A. Martins
59th Annual Meeting of the Association for Computational Linguistics (ACL)

DO CONTEXT-AWARE TRANSLATION MODELS PAY THE RIGHT ATTENTION?
K. Yin, **P. Fernandes**, D. Pruthi, A. Chaudhary, A. Martins, G. Neubig
59th Annual Meeting of the Association for Computational Linguistics (ACL)

STRUCTURED NEURAL SUMMARIZATION
P. Fernandes*, M. Allamanis, M. Brockschmidt.
7th International Conference on Learning Representations (ICLR)

WORK EXPERIENCE

FEB 2020 | Research Intern at UNBABEL, Lisbon
- AUG 2020 | *Machine Translation*
Worked on improving internal generic translations engines using latest research techniques. Rewrote the internal library to train and fine-tune models in order to bridge the gap between the experimentation and production setting.

SEPT. 2018 | AI Resident at MICROSOFT RESEARCH, Cambridge
- SEPT. 2019 | *Deep Program Understanding & Reinforcement Learning*
Worked as an AI Resident, a new 1-year training program designed to train and prepare the residents to be Researchers or Research Engineers in the AI area. Worked on unsupervised learning for anomaly detection in APIs usages in the first 6 months and worked on reinforcement learning for improving exploration in games in the last 6 months.

SUMMER 2018	Research Internship at MICROSOFT RESEARCH, Cambridge <i>Deep Program Understanding</i> Worked in trying to learn better distributed representations for source code functions using graph neural networks by exploiting control and data flow information extractable from the parse trees. Also developed a novel graph neural network model that is able to exploit a sequencing in the graph and applied this model to source code and natural language summarization
OCTOBER 2015 - JUNE 2016	Researcher at CODACY, Lisbon <i>Machine Learning for Static Code Analysis</i> Due to a good performance in the internship at INESC-ID, was invited to do some research for CODACY, a company that focus on static analysis tools for companies and developers. Our research group focus on possible applications of machine learning algorithms for automatic code analysis task where "standard" algorithms struggle achieve good results.
SUMMER 2015	Research Intern at INESC-ID, Lisbon <i>Natural Language Processing</i> Applied for a summer internship in my first year, and did some research on the application of DNA alignment algorithms to automatic audio extraction from movies, for a easier creation of labelled audio datasets to use in NLP projects

AWARDS & SCHOLARSHIPS

SEPT. 2019	Merit Scholarship for the best student of the bachelors in the year 2016/2017 (out of 300+).
SEPT. 2019	Merit Scholarship for the best student of the bachelors in the year 2015/2016 (out of 300+).
AUGU. 2016	Excellency Award for outstanding performance on the academic year 2014/2015.
SEPT. 2014	Ferreira Dias' Best Student Award of the year 2013/2014.

TALKS

APR. 2021	Explainable AI and How to Evaluate It , <i>Deep Learning Sessions Lisbon</i>
FEB. 2021	Measuring and Increasing Context Usage , <i>Reading Group, Unbabel</i>
NOV. 2018	Structured Neural Summarization , <i>AI + Pizza, Microsoft Research Cambridge</i>

SERVICE

Reviewer:

- *Language Resources and Evaluation*, 2021
- *Transactions on Software Engineering*, 2021
- *Graph Representation Learning and Beyond*, workshop at ICML 2020
- *Graph Representation Learning*, workshop at NeurIPS 2019.
- *Representation Learning on Graphs and Manifolds*, workshop at ICLR 2019.

Open-Source:

- *HuggingFace*, Library for pretrained language models, Contributor
- *OpenGNN*, Library for machine learning for graph-structured data, Sole Developer
- *scikit-learn*, Library for machine learning in Python, Minor Contributor.

SKILLS AND INTERESTS

LANGUAGES: Portuguese (Native); English(Fluent, IELTS Band 8.5); Spanish (Intermediate); Japanese (Basic)

TECHNICAL SKILLS: Python (jax, pytorch, tensorflow, scikit, numpy) , Octave, Java, C++, C

RESEARCH INTERESTS: Machine Learning, Representation Learning, Natural Language Processing, Reinforcement Learning, Program Understanding, Computational Neuroscience, Theorem Proving